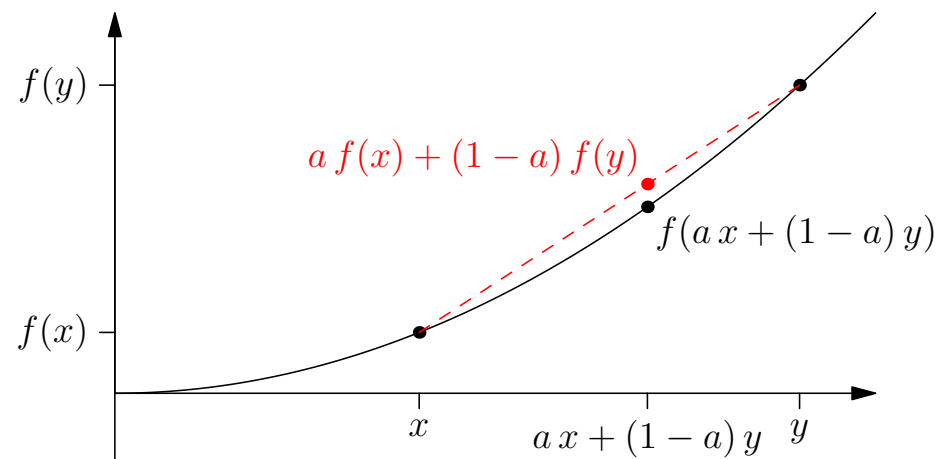


# Mathematics of Machine Learning

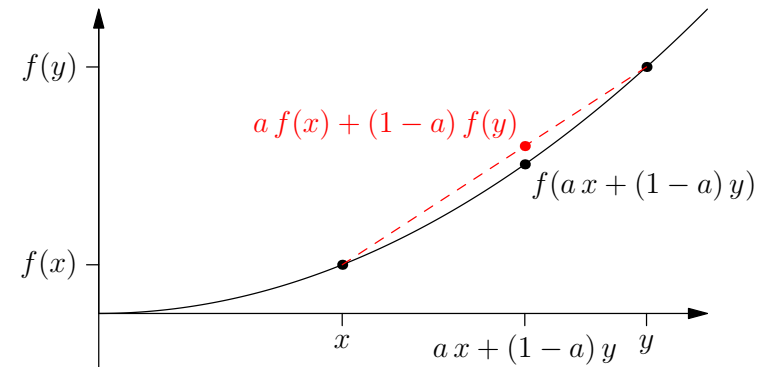
## Convexity



*Convex sets, convex functions, Jensen's inequality*

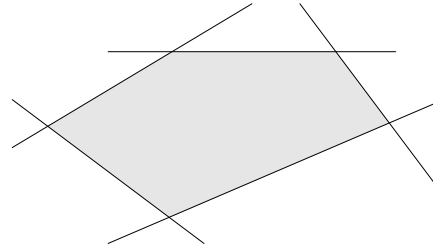
# Outline

1. **Convex sets**
2. Convex functions
3. Jensen's inequality



# Convex Regions

- Convex regions are familiar

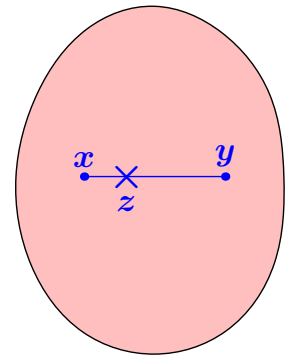


- For any two points  $x$  and  $y$  in a region  $\mathcal{R}$  then for any  $a \in [0,1]$  if

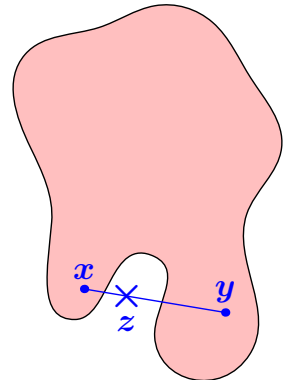
$$z = ax + (1 - a)y \in \mathcal{R}$$

- then  $\mathcal{R}$  is a convex region

Convex region



Non-convex region



# Convex Sets

- For any set,  $\mathcal{S}$ , where addition and scalar multiplication is defined (e.g. a vector space) then:

If for any two elements  $\mathbf{x}, \mathbf{y} \in \mathcal{S}$  and any  $a \in [0, 1]$

$$\mathbf{z} = a\mathbf{x} + (1 - a)\mathbf{y} \in \mathcal{S}$$

then  $\mathcal{S}$  is said to be a convex set■

# Positive Semi-Definite Matrices

- Recall that a matrix  $\mathbf{M}$  is positive semi-definite if for any vector  $\mathbf{v}$

$$\mathbf{v}^T \mathbf{M} \mathbf{v} \geq 0$$

(i.e. any quadratic form of the matrix is non-negative)

- (We showed this also implies that all the eigenvalues are non-negative)
- We denote the fact that  $\mathbf{M}$  is positive semi-definite by  $\mathbf{M} \succeq 0$ , and  $\mathbf{M} \succ 0$  if it is positive definite
- The set of positive semi-definite (PSD) matrices (or kernels) form a convex set

# Proof

- Consider any two arbitrarily chosen PSD matrices  $\mathbf{M}_1$  and  $\mathbf{M}_2$  and any  $a \in [0,1]$  then let

$$\mathbf{M}_3 = a\mathbf{M}_1 + (1 - a)\mathbf{M}_2$$

- Then for any vector  $\mathbf{v}$

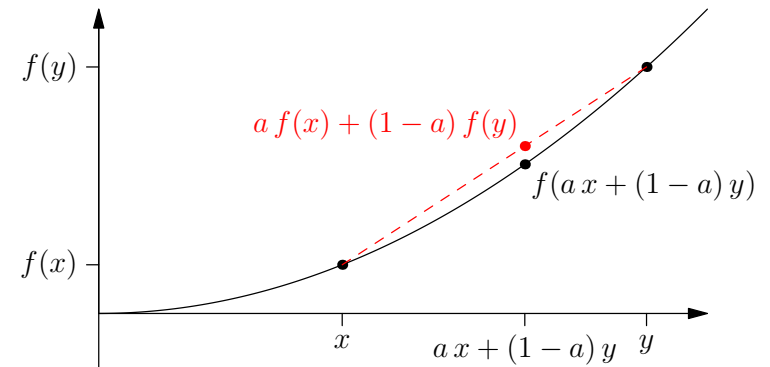
$$\begin{aligned} \mathbf{v}^\top \mathbf{M}_3 \mathbf{v} &= \mathbf{v}^\top (a\mathbf{M}_1 + (1 - a)\mathbf{M}_2) \mathbf{v} \\ &= a\mathbf{v}^\top \mathbf{M}_1 \mathbf{v} + (1 - a)\mathbf{v}^\top \mathbf{M}_2 \mathbf{v} \\ &= am_1 + (1 - a)m_2 \end{aligned}$$

where  $m_1 = \mathbf{v}^\top \mathbf{M}_1 \mathbf{v}$  and  $m_2 = \mathbf{v}^\top \mathbf{M}_2 \mathbf{v}$

- But  $m_1, m_2 \geq 0$  since  $\mathbf{M}_1, \mathbf{M}_2 \succeq 0$ . Thus  $am_1 + (1 - a)m_2 \geq 0$  and so  $\mathbf{M}_3 \succeq 0$   $\square$

# Outline

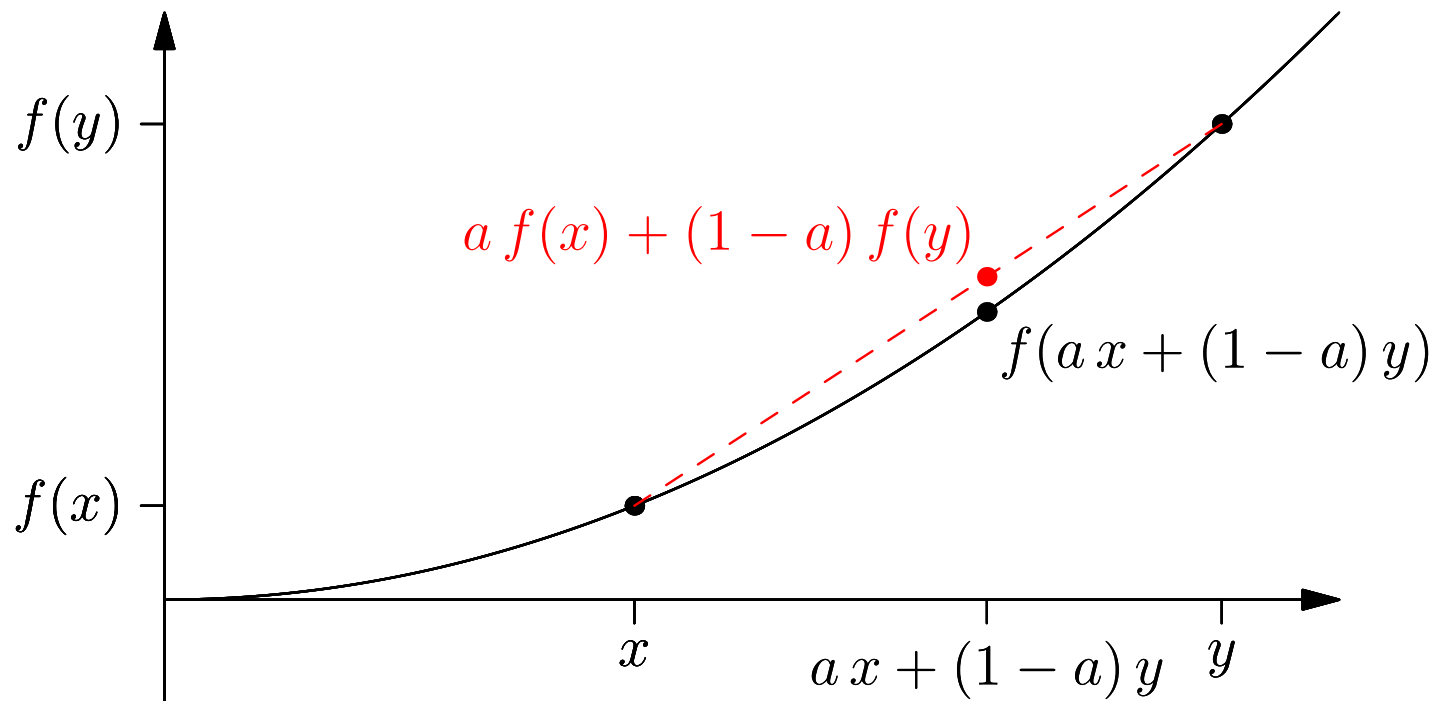
1. Convex sets
2. **Convex functions**
3. Jensen's inequality



# Convex Functions

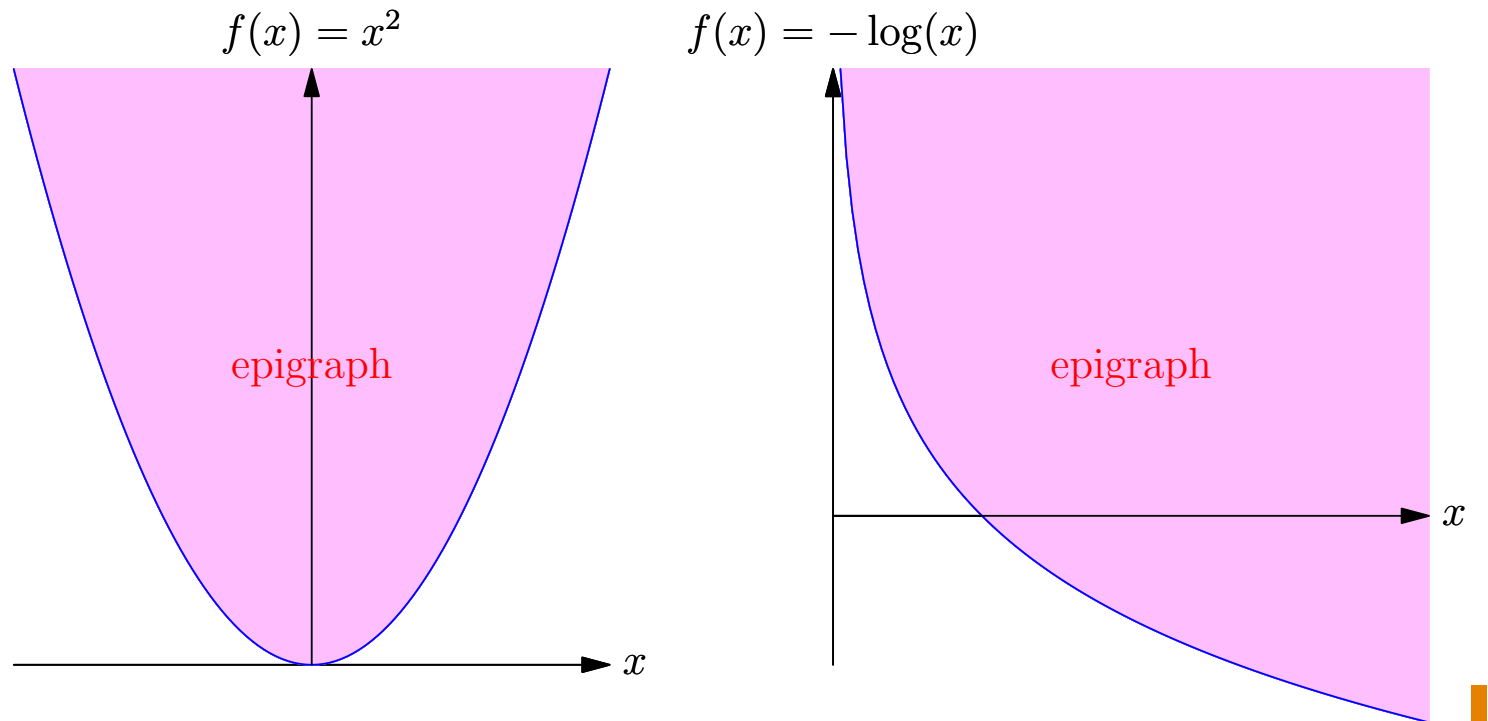
- Any function  $f(x)$  is said to be a **convex function** if for any two points  $x$  and  $y$  and any  $a \in [0,1]$

$$f(ax + (1 - a)y) \leq af(x) + (1 - a)f(y)$$



# Epigraph

- The **epigraph** of a function is the area that lies above the function■
- The epigraph of a convex function is a convex region



# Convex-Down or Concave Functions

- Any function,  $f(x)$ , that satisfies the inverse inequality

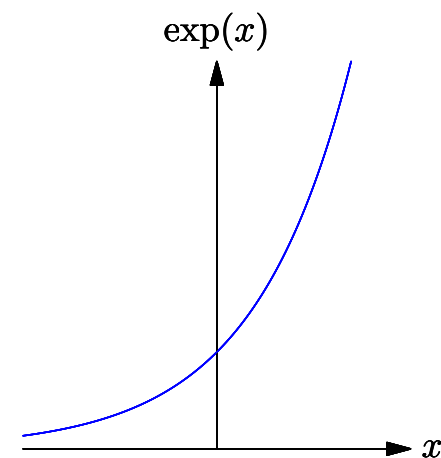
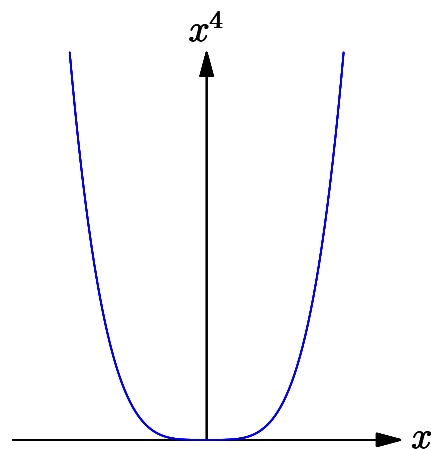
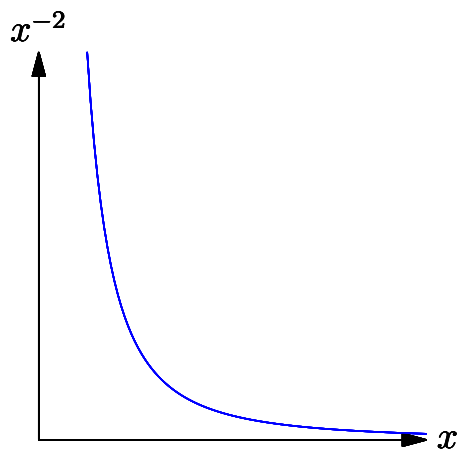
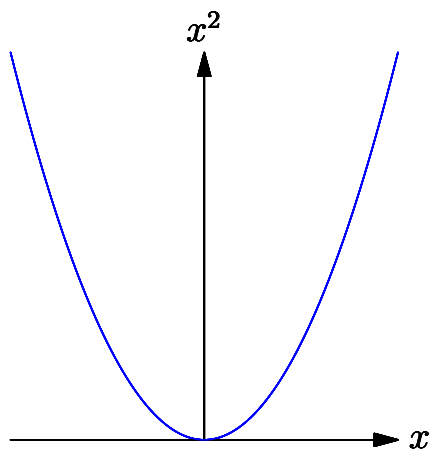
$$f(ax + (1 - a)y) \geq af(x) + (1 - a)f(y)$$

for any points  $x$  and  $y$  and any  $a \in [0,1]$  is said to be a **convex-down** or **concave** function

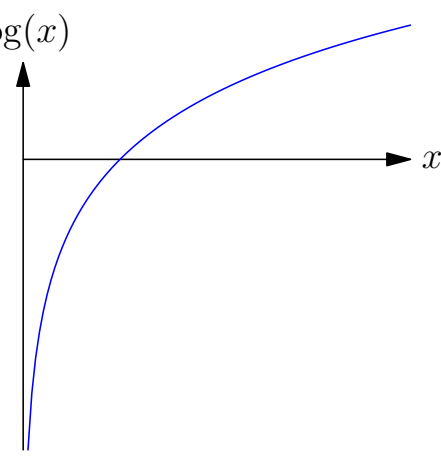
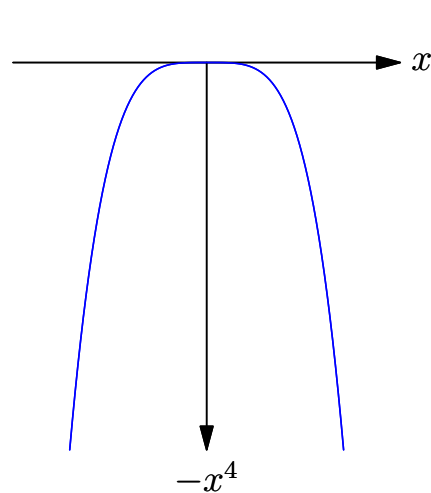
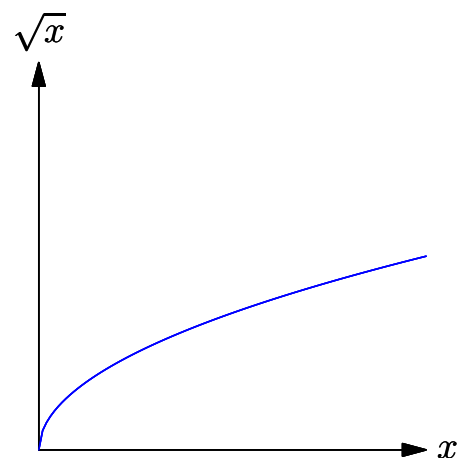
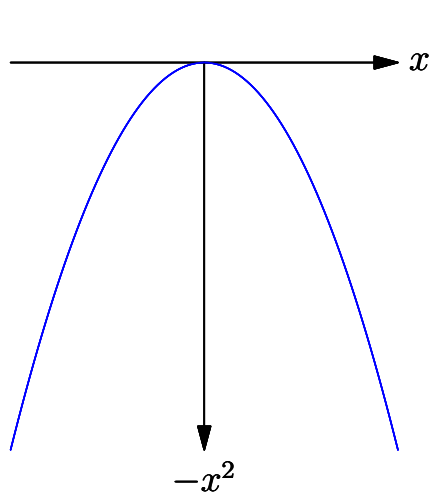
- Everything true for a convex(-up) function carries over to a convex-down function with a small modification
- If  $f(x)$  is a convex-up function then  $g(x) = -f(x)$  is a convex-down function
- The area that lies below a convex-down function is a convex region

# Examples

## Convex-Up Functions



## Convex-Down Functions



# Linear Functions

- Linear functions are given by

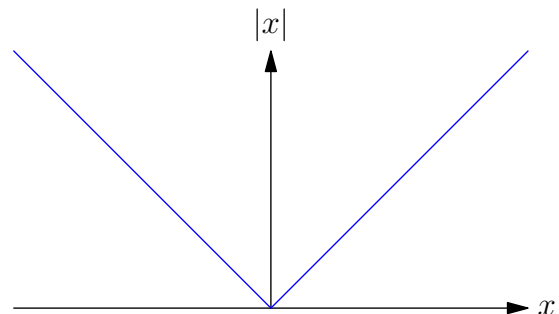
$$f(x) = mx + c$$

- They satisfy the **equality**

$$f(ax + (1 - a)y) = af(x) + (1 - a)f(y)$$

- As such they are both convex(-up) and convex-down function

- $|x|$  is a convex-up function



# Strictly Convex Function

- Functions that satisfy the strict inequality (for  $0 < a < 1$  and  $x \neq y$ )

$$f(ax + (1 - a)y) < af(x) + (1 - a)f(y)$$

are said to be **strictly convex functions**■

- A strictly convex-down function satisfies the reverse strict inequality■
- Strictly convex-(up or down) functions don't contain any linear regions■

# Convexity in High Dimensions

- If  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  (i.e.  $f(\mathbf{x})$  maps high dimensional point  $\mathbf{x} \in \mathbb{R}^n$  to a real value) satisfies

$$f(a\mathbf{x} + (1 - a)\mathbf{y}) \leq af(\mathbf{x}) + (1 - a)f(\mathbf{y})$$

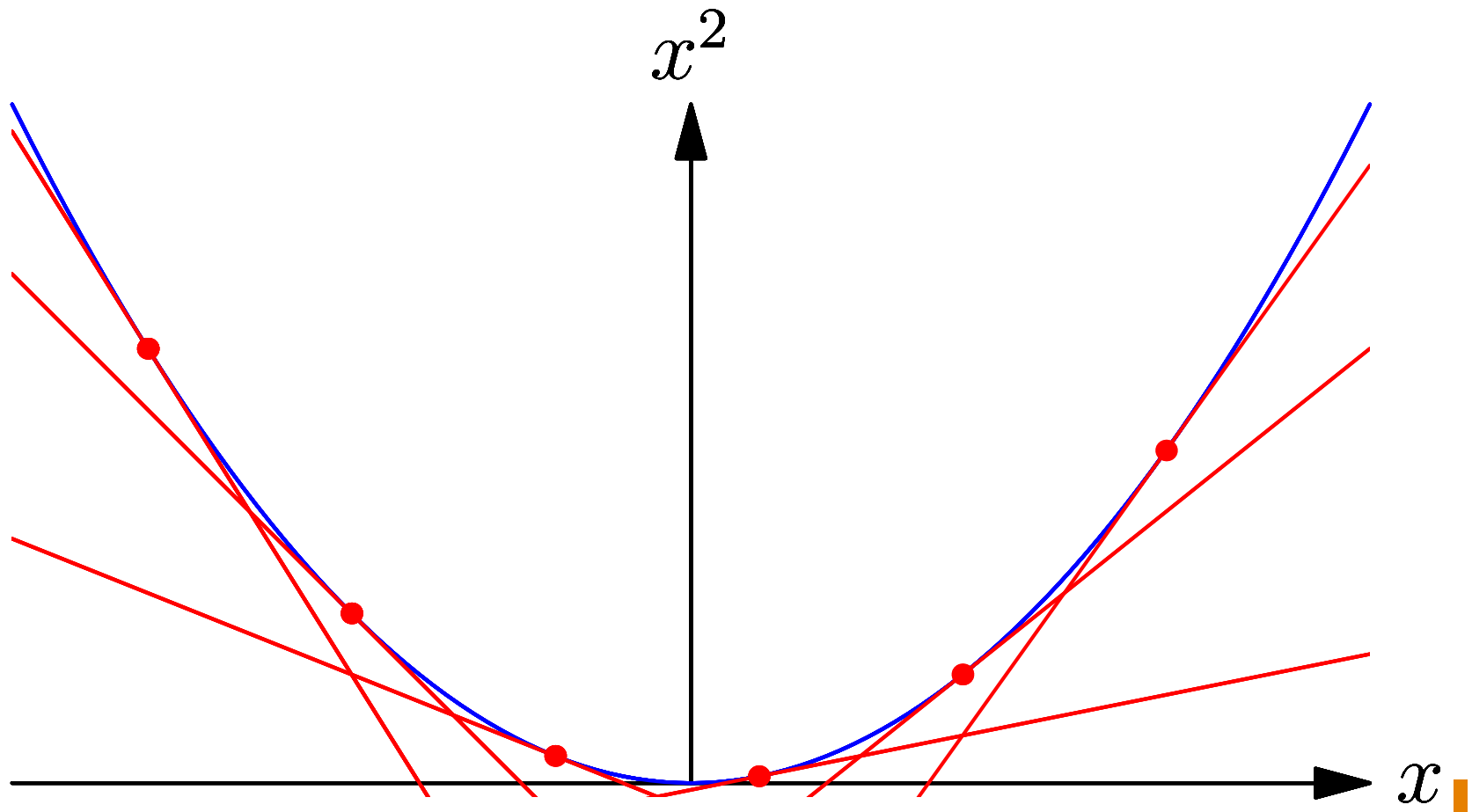
for any  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and any  $a \in [0, 1]$  then  $f(\mathbf{x})$  is a convex function■

- $\|\mathbf{x}\|_2^2 = \sum_i x_i^2$  is a (strictly) convex function■
- $\|\mathbf{x}\|_1 = \sum_i |x_i|$  is a convex function■

# Properties of Convex Functions

- Convex functions lie on or above any tangent line

$$f(x) \geq f(x^*) + (x - x^*)f'(x^*)$$



# Second Derivatives

- As  $f(x)$  lies on or above its tangent line then for any  $\epsilon > 0$

$$f'(x + \epsilon) \geq f'(x) \blacksquare$$

therefore  $f''(x) = \lim_{\epsilon \rightarrow 0} (f'(x + \epsilon) - f'(x))/\epsilon \geq 0$  at all points  $x$   $\blacksquare$

- In high dimensions a convex function lies above its tangent plane

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) \blacksquare$$

- The matrix of second derivatives (the Hessian) must be positive semi-definite at all points  $\mathbf{x}$

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \succeq 0 \blacksquare$$

# Proving Convexity

- $f(x) = x^2$  is strictly convex as  $f''(x) = 2 > 0$ ■
- $f(x) = e^{cx}$  is strictly convex as  $f''(x) = c^2 e^{cx} > 0$ ■
- $f(x) = \log(x)$  is strictly convex-down as  $f''(x) = -\frac{1}{x^2} < 0$ ■
- $f(x, y) = \frac{x^2}{y}$  is convex for  $y > 0$  as

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 f(x, y)}{\partial x^2} & \frac{\partial^2 f(x, y)}{\partial x \partial y} \\ \frac{\partial^2 f(x, y)}{\partial y \partial x} & \frac{\partial^2 f(x, y)}{\partial y^2} \end{pmatrix} = \begin{pmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{pmatrix} = \frac{2}{y^3} \begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix} \blacksquare$$

- Now  $T = \text{tr} \mathbf{H} = \frac{2}{y^3}(x^2 + y^2)$ ,  $D = \det(\mathbf{H}) = 0$ ■
- $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D} = \{0, T\} = \{0, \frac{2(x^2 + y^2)}{y^3}\} \geq 0 \Rightarrow \mathbf{H} \succeq 0$ ■

# Sums of Convex Functions

- For any set of convex functions  $f_1(x), f_2(x), \dots$  and any set of non-negative scalars  $a_1, a_2, \dots$  then

$$g(x) = \sum_i a_i f_i(x)$$

is convex■

- Proof

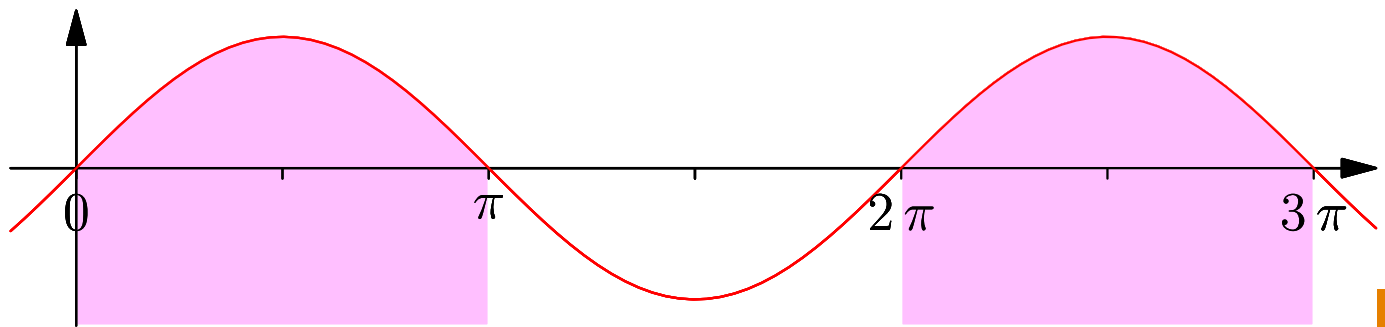
$$g''(x) = \sum_i a_i f_i''(x)$$

but  $f_i''(x) \geq 0$  so  $g''(x)$  is a sum on non-negative terms■

- This generalises to higher dimensions as the sum of PSD matrices times positive factors is a PSD matrix■

# Convex Functions Defined on Convex Sets

- All the properties we have discussed hold for functions defined on a convex set■
- $\sin(x)$  is not generally neither convex up or down■
- $\sin(x)$  for  $x \in [0, \pi]$  is convex-down■ (its second derivative  $-\sin(x)$  is less than or equal to 0 in this range)■



- For a convex function defined on a non-convex set,  $\mathcal{S}$ , there exists points  $x, y \in \mathcal{S}$  such that for some  $a \in [0, 1]$  there will be points  $z = ax + (1 - a)y \notin \mathcal{S}$ ■ (the function isn't defined on such points)■

# Constraints

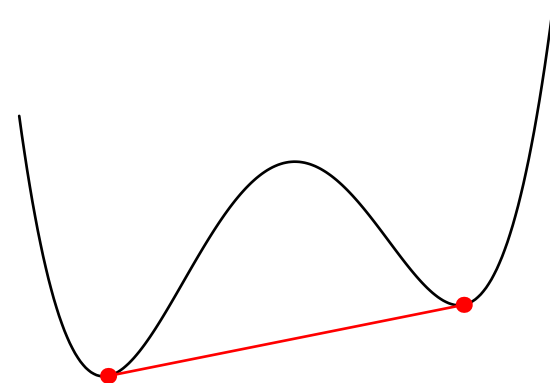
- Often we impose constraints on the set of points, e.g.

$$x_i > 0 \qquad \mathbf{a}^\top \mathbf{x} = b \qquad \mathbf{x}^\top \mathbf{M} \mathbf{x} \leq 1$$

- Linear constraints (e.g.  $x_i > 0$  or  $\mathbf{a}^\top \mathbf{x} = b$  or  $\mathbf{a}^\top \mathbf{x} \leq b$ ) always define a convex region■
- Multiple linear constraints always define a convex region■
- Non-linear constraints may or may not define a convex region■  
( $\{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x}^\top \mathbf{M} \mathbf{x} \leq 1, \mathbf{M} \succeq 0\}$  does while  
 $\{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x}^\top \mathbf{M} \mathbf{x} \geq 1, \mathbf{M} \succeq 0\}$  doesn't)■

# Unique Minimum

- Strictly convex function have a unique minimum■
- The existence of a local minimum would break convexity■
  - ★ The line connecting a local minimum to a global minimum would be strictly decreasing■
  - ★ Thus there are points next to the local minimum with lower values■
  - ★ This is a contradiction■
- This remains true if we consider convex functions that are constrained to live in a convex set■



# Convex Set of Minima

- If  $f(x)$  is **convex** but not **strictly convex** then there might exist a convex set  $\mathcal{M} \subset \mathcal{X}$  of minima such that for all  $x, y \in \mathcal{M}$  and any  $z \in \mathcal{X}$  we have  $f(x) = f(y) \leq f(z)$ ■
- This set of minima is convex, that is, if  $x, y \in \mathcal{M}$  then for any  $a \in [0, 1]$  the point  $z = ax + (1 - a)y \in \mathcal{M}$ ■
- The sum of a convex function,  $f(x)$ , and a strictly convex function  $g(x)$  will always be strictly convex since

$$f''(x) + g''(x) > 0$$

as  $f''(x) \geq 0$  and  $g''(x) > 0$ ■

# Linear Regression

- For linear regression the loss function

$$L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 = \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}$$

is convex■

- Since the Hessian  $\mathbf{H} = 2\mathbf{X}^\top \mathbf{X} \succeq 0$  (positive semi-definite)■  
(For any vector  $\mathbf{v}$  then  $\mathbf{v}^\top \mathbf{H} \mathbf{v} = 2\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = 2\|\mathbf{X}\mathbf{v}\|^2 \geq 0$ )■
- If  $\mathbf{H} \succ 0$  there will be a unique minima■, while if  $\mathbf{H}$  has some zero eigenvalues there will be a family of solutions■

# Regularised Linear Regression

- In ridge regression we minimise a loss

$$L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \eta\|\mathbf{w}\|^2 = \mathbf{w}^\top (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}$$

- Because  $\|\mathbf{w}\|^2$  is strictly convex the loss function is strictly convex and so will have a unique solution
- Using an  $L_1$  regulariser (Lasso)

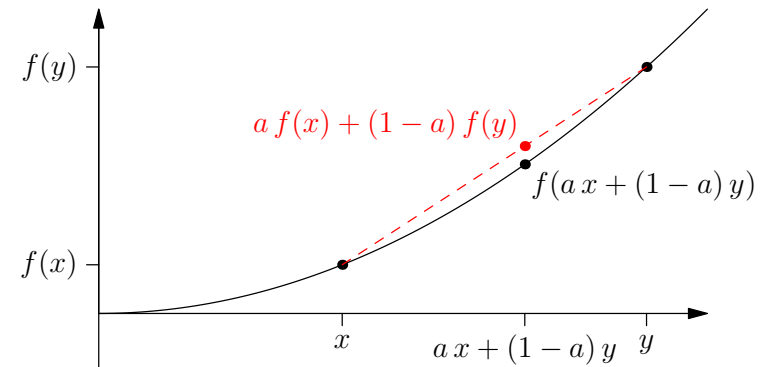
$$L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \eta\|\mathbf{w}\|_1$$

again  $\|\mathbf{w}\|_1$  is convex and so  $L(\mathbf{w})$  will be convex

- Using an  $L_1$  and an  $L_2$  regulariser (elastic net) also gives a unique solution

# Outline

1. Convex sets
2. Convex functions
3. **Jensen's inequality**



# Jensen's Inequality

- In proving many properties of learning machines inequalities are really useful■
- One of the most useful inequalities involve expectations of convex functions, this is known as **Jensen's Inequality**■
- If  $f(x)$  is a convex(-up) function then

$$\mathbb{E}[f(\mathbf{X})] \geq f(\mathbb{E}[\mathbf{X}])■$$

- If  $f(x)$  is a concave(-down) function then

$$\mathbb{E}[f(\mathbf{X})] \leq f(\mathbb{E}[\mathbf{X}])■$$

# Proof

- We said before that a convex function must lie on or above its tangent plane at any point  $\mathbf{x}^*$

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) \blacksquare$$

- Taking  $\mathbf{x}^* = \mathbb{E}[\mathbf{X}]$

$$f(\mathbf{X}) \geq f(\mathbb{E}[\mathbf{X}]) + (\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top \nabla f(\mathbb{E}[\mathbf{X}]) \blacksquare$$

- Taking expectations of both sides

$$\begin{aligned} \mathbb{E}[f(\mathbf{X})] &\geq f(\mathbb{E}[\mathbf{X}]) + (\mathbb{E}[\mathbf{X}] - \mathbb{E}[\mathbf{X}])^\top \nabla f(\mathbb{E}[\mathbf{X}]) \blacksquare \\ &= f(\mathbb{E}[\mathbf{X}]) \quad \square \blacksquare \end{aligned}$$

# Simple Proofs with Jensen's Inequality

- Since  $f(x) = x^2$  is convex by Jensen's inequality

$$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2 \quad \text{or} \quad \mathbb{E}[X^2] - \mathbb{E}[X]^2 \geq 0$$

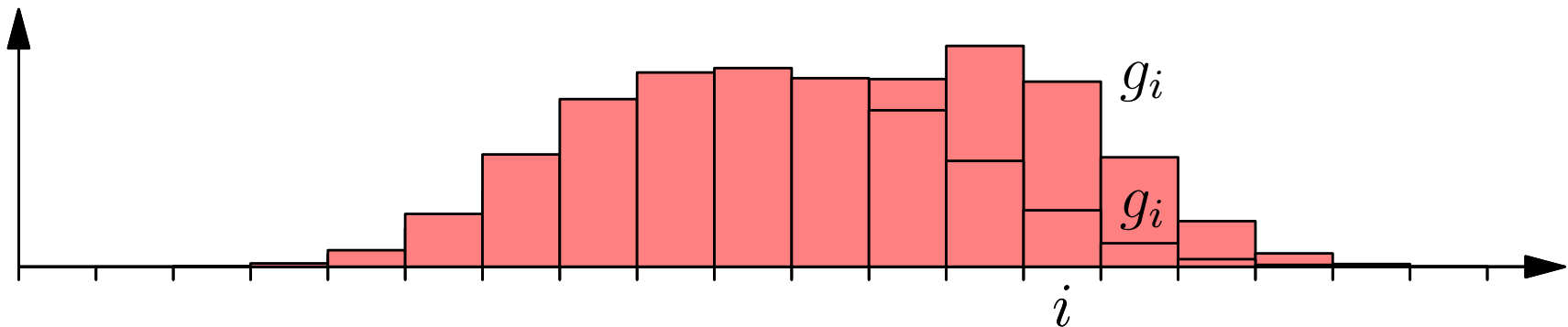
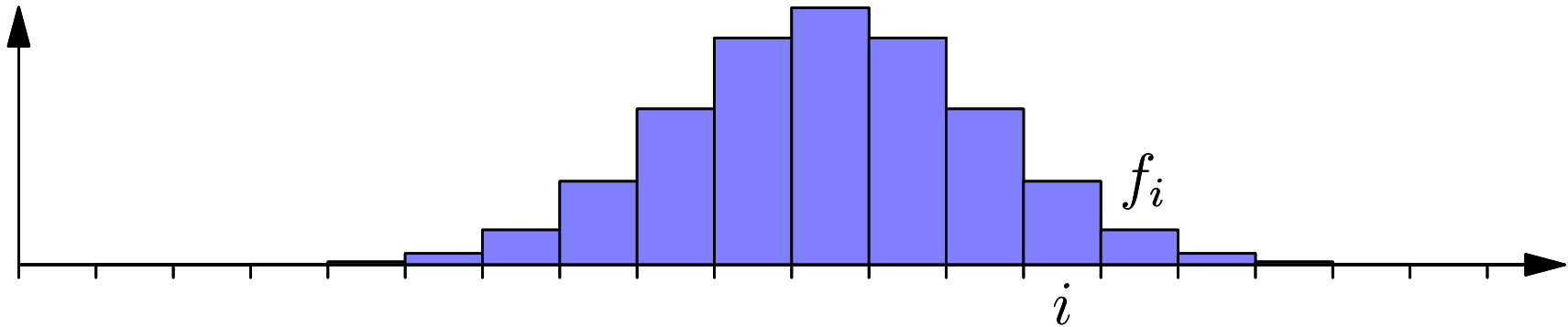
(i.e. variance are non-negative) ■

- The KL-divergence  $\text{KL}(f\|g)$  between two categorical probability distributions  $(f_1, f_2, \dots)$  and  $(g_1, g_2, \dots)$  is define as

$$\text{KL}(f\|g) = - \sum_i f_i \log \left( \frac{g_i}{f_i} \right)$$

(note  $f_i, g_i \geq 0$  and  $\sum_i f_i = \sum_i g_i = 1$ ) ■

# Kullback-Leibler Divergence



$$\text{KL}(\mathbf{f} \parallel \mathbf{g}) = - \sum_{i=1}^n f_i \log \left( \frac{g_i}{f_i} \right) = 0.237$$

# Proof of Gibbs' Inequality

- To show that  $\text{KL}(f\|g) \geq 0$  (Gibbs' inequality) we note that since the logarithm is a convex-down function

$$\begin{aligned}\text{KL}(f\|g) &= -\sum_i f_i \log\left(\frac{g_i}{f_i}\right) = \mathbb{E}_f \left[ -\log\left(\frac{g_i}{f_i}\right) \right] \\ &\geq -\log\left(\mathbb{E}_f \left[ \frac{g_i}{f_i} \right]\right) \\ &= -\log\left(\sum_i f_i \frac{g_i}{f_i}\right) = -\log\left(\sum_i g_i\right) = -\log(1) = 0\end{aligned}$$

- We will meet KL-divergences later on

# Lessons

- Although we haven't talked much about machine learning, convexity is heavily used in many machine learning applications■
- A lot of ML algorithms involve convex functions■e.g. SVMs■
- As such they will have a unique minimum (or a convex set of minima)■
- Convexity is an elegant idea which is relatively easy to prove theorems about■
- One of the most useful tools is Jensen's inequality■